Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

# SDC Hackathon

# Technical Tools Declaration

**Hosted by Xiamen University Malaysia (XMUM) AI Club**

**Submission Deadline: 25 October 2026**

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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**SDC HACKATHON 2026**

# Technical & Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club
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## 1. Team & Project Overview

| | |
|---|---|
| **Team Name** | KQ BuildLab |
| **Project / Solution Name** | FutureNow |
| **Name (Leader)** | Lai Kai Qi |
| **Email & Phone (Leader)** | kaiqi1004@gmail.com |
| **University Affiliation(s)** | UNIMAS |
| **Live Prototype URL** | https://future-now.replit.app |
| **Code Repository URL (public / access granted to judges)** | kaiqi1004-star/FutureNow |

## 2. Architecture & Technology Stack

### 2.1 System Architecture Overview

*Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.*

***[Maximum 200 words]***

FutureNow is a web-based career exploration and workforce insights prototype. Its codebase uses a pnpm workspace with a TypeScript-based frontend and an Express 5 API server. The project configuration also identifies PostgreSQL and Drizzle ORM for database access.

The application is designed to support skills-profile assessment, career matching and Malaysian labour-market insights. The documented architecture keeps user skills profiles in browser-local storage and separates supplied job-posting records from national labour-market statistics. User profiles are not intended to be stored in a backend account system under the current architecture.

The actual data sources, AI services, database connectivity and deployment configuration must be declared according to the running implementation. The public repository is available at https://github.com/kaiqi1004-star/FutureNow.

### 2.2 Languages, Frameworks & Libraries

*List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.*

| Name & Version | Purpose in Project | License |
|---|---|---|
| TypeScript 5.9.x | Application and server code | Apache-2.0 |
| Node.js 24 | JavaScript runtime | MIT |
| Express 5 | API server | MIT |
| PostgreSQL | Relational database, if connected in the submitted deployment | PostgreSQL License |
| Drizzle ORM | Database access and schema management | Apache-2.0 |
| pnpm | Workspace and package management | MIT |

| Name & Version | Purpose in Project | License |
| --- | --- | --- |
| Zod | Data validation, if included in the shipped application | MIT |

## 3. APIs & Cloud Services

### 3.1 External APIs

*List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.*

| API & Provider | Purpose in Project | Access Tier (free / paid / sponsor-provided) |
|---|---|---|
| | | |
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| | | |
| | | |

### 3.2 Cloud Platforms & Deployment

*List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.*

| Service & Provider | Purpose in Project | Plan / Region |
|---|---|---|
| Replit | Required development platform and prototype hosting, subject to the actual deployment configuration | Current account plan and deployment region: verify in Replit |
| PostgreSQL | Database technology declared in the repository configuration; include as an active cloud database only if the submitted deployment connects to one | Hosting provider, plan and region: verify |
| GitHub | Public source-code repository | Public repository; hosting region not applicable |
| | | |
| | | |

## 4. AI Tools, Models & Datasets

### 4.1 AI Models Used in the Solution

*List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.*

| Model & Version | Source / Provider | Usage (pre-trained / fine-tuned / self-trained) | License / Terms |
|---|---|---|---|
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### 4.2 Datasets

*List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.*

| Dataset Name | Source / URL | Type (public / private / self-collected / synthetic) | License | Personal Data? (Y/N) |
|---|---|---|---|---|
| Labour Force Statistics, August 2026 | DOSM | Public | Government Open Data Terms of Use 1.0, subject to source confirmation | N |
| Youth Unemployment, June 2026 | DOSM | Public | Government Open Data Terms of Use 1.0, subject to source confirmation | N |

| Dataset Name | Source / URL | Type (public / private / self-collected / synthetic) | License | Personal Data? (Y/N) |
|---|---|---|---|---|
| Skills-Related Underemployment by Age, Q2 2026 | DOSM | Public | Government Open Data Terms of Use 1.0, subject to source confirmation | N |
| Annual Salaries & Wages by Skill Category and Sex, 2025 | DOSM | Public | Government Open Data Terms of Use 1.0, subject to source confirmation | N |
| MYFutureJobs job-posting records collected for the project | MYFutureJobs | Public | MYFutureJobs Terms and Conditions; permission for reuse and redistribution not confirmed | N |

### 4.3 AI-Assisted Development Tools

*Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.*

Did your team use any generative-AI tool to help create code, design, text, or other project material?

/ Yes ☐ No

If yes, list them below:

| **Tool & Version** | **Used For** | **Extent of Use (minor / moderate / major)** |
|---|---|---|
| ChatGPT | Coding assistance, debugging, prompt drafting and documentation | Major |
| Replit | Development, running and deployment workflow | Major |
| | | |
| | | |
| | | |

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

***[Maximum 100 words]***

The team reviews AI-assisted code and content against the project requirements and available source data. Dataset values are checked against their original sources, while application features are tested using representative inputs and expected outputs. Matching scores and scenario results are presented as estimates rather than guarantees. The team remains responsible for verifying factual claims, testing functionality, checking dependencies and licences, and correcting identified errors before submission.

## 5. Sponsor-Provided Resources

*Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.*

Did your project use any official SDC sponsor-provided resource?

/ Yes ☐ No

If yes, list them below:

| **Sponsor** | **Resource (API / credits / platform / data)** | **How It Was Used in the Project** |
|---|---|---|
| Replit, provided through the official SDC hackathon arrangement | Replit Core access | Used for developing, running, testing and potentially deploying the FutureNow prototype. |
| | | |
| | | |
| | | |

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

***[Maximum 100 words]***

Replit Core supported the development and testing workflow for FutureNow. The application's core career-matching functionality is determined by its implemented code and data, rather than by the subscription itself. However, Replit is the required development platform for the hackathon, so the team must comply with the event's platform requirements.

## 6. Security, Privacy & Responsible Use

### 6.1 Data Handling & Privacy

*What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.*

***[Maximum 150 words]***

FutureNow uses labour-market statistics and job-posting records to support career matching and workforce insights. The application may process user-provided skills, education, experience and career preferences. CV upload, persistent storage and third-party AI processing will be documented according to the features actually implemented. The team will minimise collected information, avoid including unnecessary personal identifiers in job datasets, and keep API credentials out of public source code. Any data transmitted to third-party services, storage locations, retention periods and deletion mechanisms will be disclosed accurately before submission. If the prototype does not store user profiles or collect personal data beyond temporary inputs, this will be stated explicitly. No claim of encryption, automatic deletion or secure storage will be made unless verified in the implementation.

### 6.2 Known Limitations & Risks

*Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.*

***[Maximum 150 words]***

FutureNow's recommendations depend on the completeness, representativeness and recency of its available job records and labour-market data. A limited collection of job postings cannot represent the entire Malaysian labour market or establish national hiring demand. Skills matching may overlook equivalent skill names or transferable experience, while AI-generated explanations may contain inaccuracies if generative AI is used. Future workforce scenarios are estimates, not guaranteed predictions. The application will distinguish official statistics from collected job-posting snapshots, display source dates where available, explain matching factors, and communicate uncertainty. Recommendations should support, rather than replace, users' own career decisions. Further validation is required to assess matching quality, dataset coverage, bias and performance across different user profiles.

## 7. Compliance & Readiness Checklist

*Tick all boxes that apply before submitting.*

/ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.

/ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.

/ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.

/ Any use of generative-AI tools for development has been disclosed and reviewed by the team.

/ Any use of official SDC sponsor resources has been declared in Section 5.

/ The prototype described here is accessible to judges through the links provided in Section 1.

/ All fields in this template have been completed within the word limits stated.

## 8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.



[Lai Kai Qi]

25/10/2026